

$$0 = 1 - 1 = -1 + 1 = 0$$

From Elementary Arithmetic to Intrinsic Skein-Lasagna

Keyao Peng

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Question

Can you prove $0 = 0$, non-trivially?

Test your math level

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$$S^3 \sim \mathbb{C}^2 \setminus \{0\} \rightarrow \mathbb{CP}^1 \sim S^2$$



$$\pi_4(S^3) \cong \mathbb{Z}/2\mathbb{Z}$$

$$\pi_1^s(S) \cong \Omega_1^{\text{fr}}$$

$$K_1(\mathbb{F}_1) \cong S_\infty^{\text{ab}}$$

$$0=1-1=1+1=0$$

imgflip.com



Hopf fibration is a map defined by

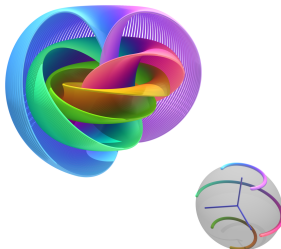
$$H : \mathbb{S}^3 \subset \mathbb{C}^2 - \{0\} \rightarrow \mathbb{CP}^1 \cong \mathbb{S}^2, (z_1, z_2) \mapsto [z_1, z_2]$$

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For any point $x \in \mathbb{S}^2$, preimage $H^{-1}(x)$ is a circle $\mathbb{S}^1$. We can visualize this map by considering $H : \mathbb{R}^3 \cong \mathbb{S}^3 - \{\infty\} \rightarrow \mathbb{S}^2$:



Homotopy groups

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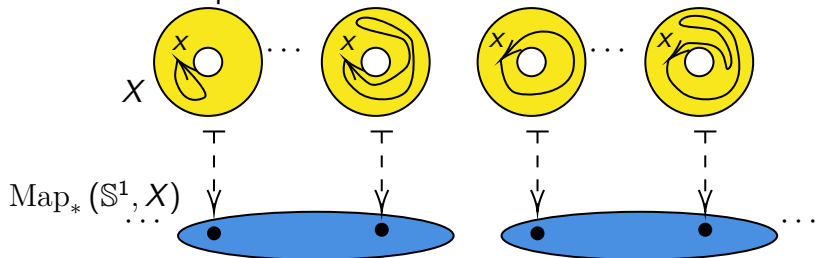
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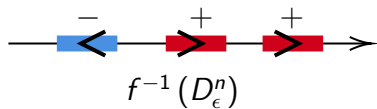
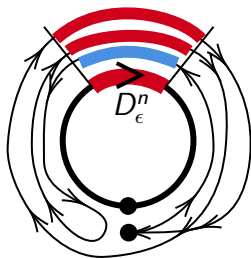
Let us study the cases $X = \mathbb{S}^n$.

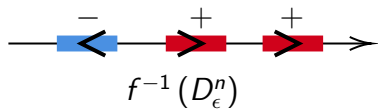
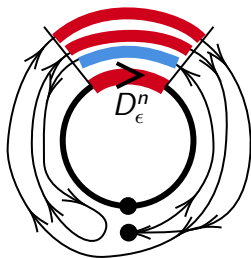
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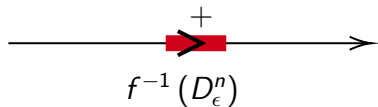
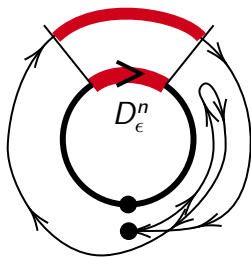
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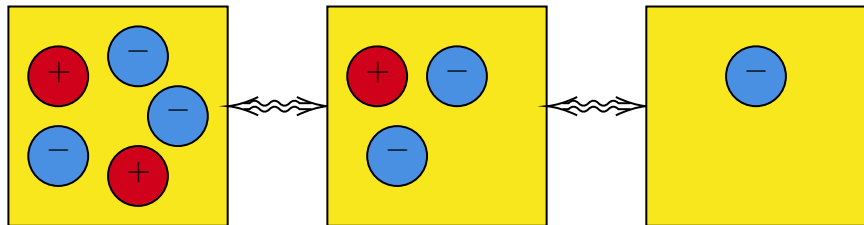
deform



Homotopy groups of spheres

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This gives an isomorphism to the configuration space of disks.

$$\text{End}_*(\mathbb{S}^n) \cong \text{Conf}^{fr}(\mathbb{R}^n) = \bigcup_{j,k} \text{Emb}(\bigsqcup_j D_+^n \sqcup \bigsqcup_k D_-^n, \mathbb{R}^n)$$

Therefore, $\pi_0 \text{End}_*(\mathbb{S}^n) \cong \mathbb{Z}$, $(j, k) \mapsto j - k$

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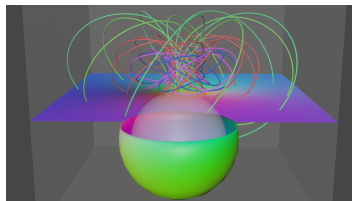
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$$\text{Map}_*(\mathbb{S}^{n+1}, \mathbb{S}^n) \cong \text{Map}_*(\mathbb{S}^1, \text{End}_*(\mathbb{S}^n)) \rightarrow \text{Map}_*([0, 1], \text{End}_*(\mathbb{S}^n))$$

Then H corresponds to a path (deformation or homotopy) from $0 \in \text{End}_*(\mathbb{S}^2)$ to 0 itself.

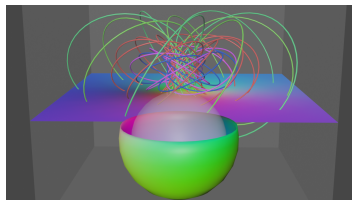
Visualize H as deformation

I made an animation to visualize this deformation.



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We can also visualize it as a movement in the configuration space.
We will show it in the end.

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We use the **Grothendieck group** $K_0(\text{FinSet})$:

$$K_0(\text{FinSet}) = \{(X, Y) \in (\text{FinSet}_{/\cong})^2\} / \{(X \sqcup Z, Y \sqcup Z) \sim (X, Y)\}$$

We can think we formally define " $X - Y$ " in this way.

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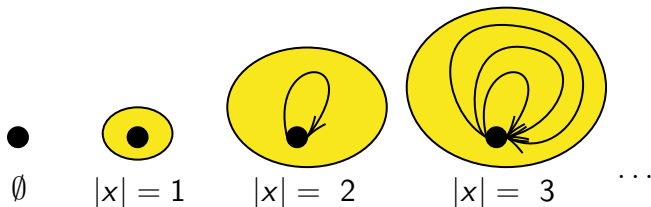
But what is $K_i(\text{FinSet})$ for $i > 0$?

Space of "finite sets"

Instead of thinking the set of "finite sets", we now consider the space of them $F = \text{FinSet}^{\cong}$: every point $x \in F$ corresponds to a finite set, and a path between two points corresponds to an isomorphism $f : x \xrightarrow{\cong} y$ of two finite sets.

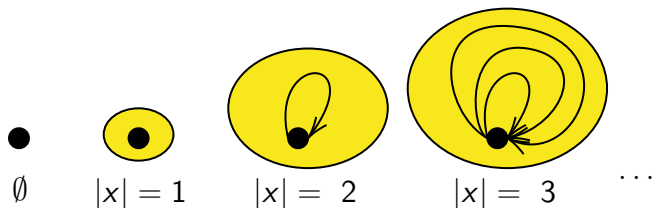
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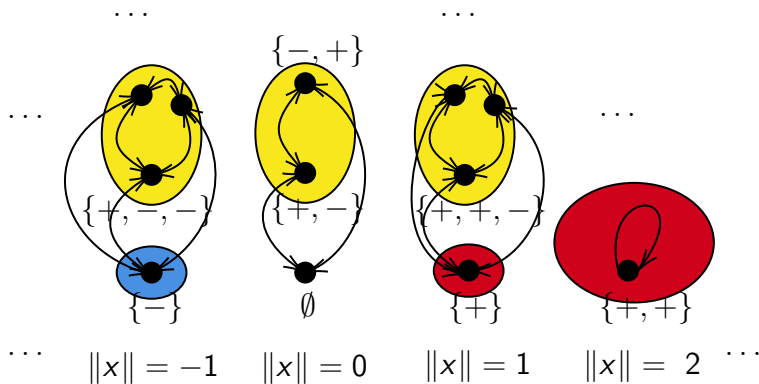
- $\pi_0(F) = \text{FinSet}_{/\cong} \cong \mathbb{N}$.
- $\pi_1(F, x) = S_{|x|}$, the permutation group.
- $\pi_i(F, x) = \{0\}$ for $i > 1$ (one can never deform an isomorphism of set to another).

K-space (spectrum) of "finite sets"

We want to add "negative" to F . We can consider the space $K(F)$ of (ordered) finite sets, whose elements are marked by $+$ or $-$, and we also want to identify the sets with same "value" $\|X\| = |X_+| - |X_-|$, and these identifications will be the paths.

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Answer

We can prove $0 = 0$ non-trivially with $h : 0 = 1 - 1 = -1 + 1 = 0$!

Algebraic K-groups of a field

Let k be a field, we can follow the same construction for finite dimensional k vector space. Let $V_k = \text{FinVect}_k^{\cong}$, we define **Algebraic K-groups** of k as $K_i(k) = \pi_i K(V_k)$.
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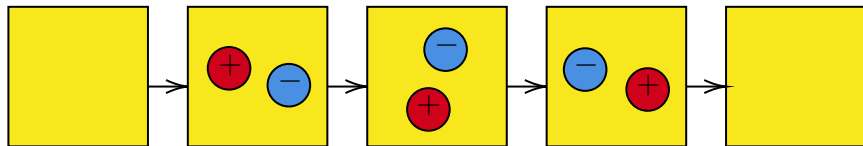
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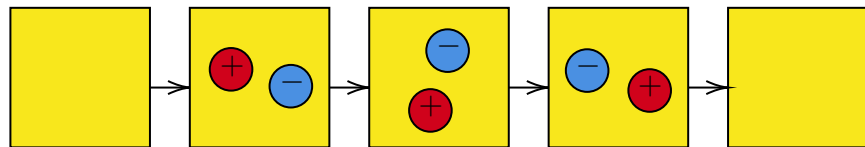


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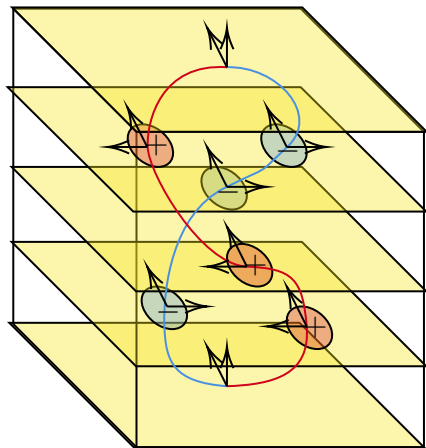


New Question

Can we find geometric interpretation for general K-groups $K_i(k)$?

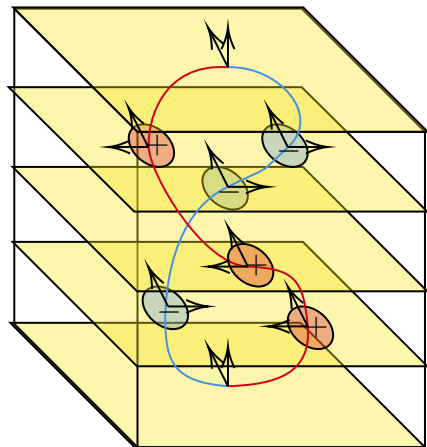
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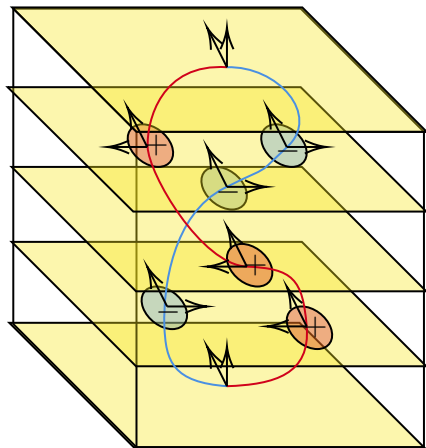
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This provides a framed circle, meaning an embedding $S^1 \hookrightarrow \mathbb{R}^n$ with a **trivialization** τ of the normal bundle. We denote Ω_i^{fr} as the group of cobordism classes of i -dimensional framed manifolds.

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This provides a framed circle, meaning an embedding $S^1 \hookrightarrow \mathbb{R}^n$ with a **trivialization** τ of the normal bundle. We denote Ω_i^{fr} as the group of cobordism classes of i -dimensional framed manifolds. And we also have:

$$\Omega_1^{fr} \cong \pi_1^s(\mathbb{S})$$

Therefore, this framed circle is non-trivial and is the generator of Ω_1^{fr} .

A n -**treefold** is a space generally locally homeomorphic to $T \times \mathbb{R}^{n-1}$, where T is a tree.

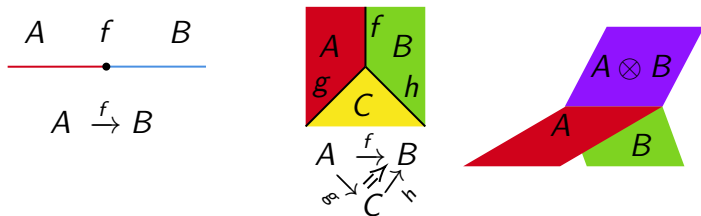
A n -**treefold** is a space generally locally homeomorphic to $\mathcal{T} \times \mathbb{R}^{n-1}$, where $\mathcal{T}$ is a tree. Give a symmetric monoidal category $(\mathcal{C}, \otimes)$, define an n - $(\mathcal{C}, \otimes)$ -**lasagna** L as a framed n -treeifold $\mathcal{T}$ with a stratification $\mathcal{S}$ and the strata are marked compatibly with category $\mathcal{C}$:

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- 0-codimensional strata are marked with objects; For $i > 0$, i -codimensional strata are marked with i -morphism,
- At the branches of treefold marked with the monoidal product $\otimes$.

Example



Relation with "Skein-lasagna module"

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We can think a lasagna L as a string diagram of $\mathcal{C}$ drawing on $\mathcal{T}$, or another way to say $L \in \int_{\mathcal{T}}(\mathcal{C}, \otimes)$.

Cobordism of lasagna

As one expects, we can define the cobordism relation between L_1, L_2 , if there is a lasagna W with boundary such that $\partial W = L_1 \sqcup -L_2$.

Finally, we define the n -th $(\mathcal{C}, \otimes)$ -**cobordism group** $\Omega_n^{\mathcal{C}, \otimes}$ to be the abelian group generated by n -lasagna modulo cobordism relations.

$(V_k, \oplus)$ -Cobodism Group

As an example $\Omega_n^{V_k, \oplus}$, where $V_k = \text{FinVect}_k^{\cong}$ is the subcategory of finite-dimensional vector space where the morphisms are all isomorphisms.

$(V_k, \oplus)$ -Cobodrisim Group

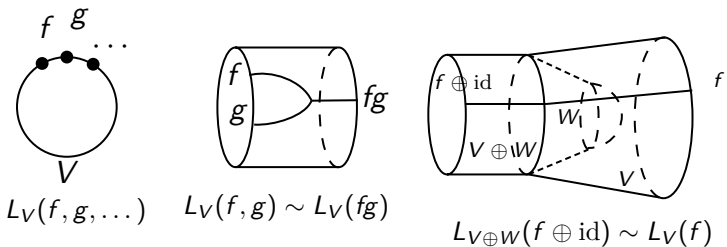
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- It is clear $\Omega_0^{V_k, \oplus} \cong \mathbb{Z}$ is classified dimension.

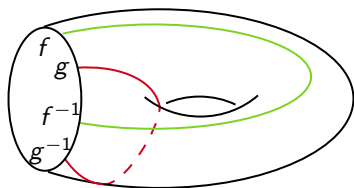
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- It is clear $\Omega_0^{V_k, \oplus} \cong \mathbb{Z}$ is classified dimension.
- $\Omega_1^{V_k, \oplus}$ is generated by circle $L_V(f)$ marked with vector spaces V and matrix $f \in GL_V(k)$. And by adding identity matrices, we find we have a homomorphism $L : GL_\infty(k) \rightarrow \Omega_1^{V_k, \oplus}$.

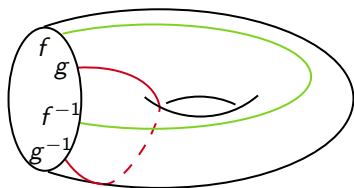


We have one more relation given by surfaces:



$$L([f, g]) \sim 0$$

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Notice that $k^\times \cong \mathrm{GL}_\infty(k)/[\mathrm{GL}_\infty(k), \mathrm{GL}_\infty(k)]$. In fact, one can show the induced homomorphism $L : k^\times \xrightarrow{\cong} \Omega_1^{V_k, \oplus}$ is isomorphism. And the lasagna induced by Hopf map H is given by $L(-1)$.

Generalized Pontryagin–Thom Isomorphism

Recall that $K_0(k) = \mathbb{Z}$, $K_1(k) = k^\times$. This is not just a coincidence:

Generalized Pontryagin–Thom Isomorphism (My claim)

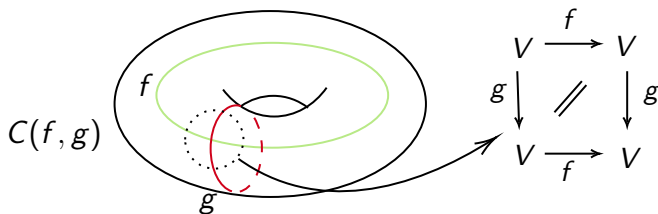
For all $n \geq 0$, we have the isomorphism

$$K_n(k) \xrightarrow{\cong} \Omega_n^{V_k, \oplus}$$

This answers the new question, which indicates we can view lasagna as elements in K -groups, which provides a bridge from hard problems in algebra toward hard problems in geometric topology.

So what does it tell us about $n = 2$?

For matrices $f, g \in \mathrm{GL}(k)$ such that $[f, g] = 1$, one can define following 2-lasagna $C(f, g)$:



Recall $K_2(k) = k^\times \otimes k^\times / \langle a \otimes 1 - a \rangle$, which means we can find a 3-lasagna who bounds $C(a, 1 - a)$. In other word, we can encode prove/computation by a lasagna.

I don't have a lasagna bounds $C(a, 1 - a)$ for now, but I have an easier example.

3-Lasagna bounds $C(e_{12}(a), e_{12}(b))$

For $i \neq j$, let elementary matrices $e_{ij}(a)$ to be

$$(e_{ij}(a))_{kl} = \begin{cases} 1, & \text{if } k = l, \\ a, & \text{if } k = i, l = j, \\ 0, & \text{otherwise.} \end{cases}$$

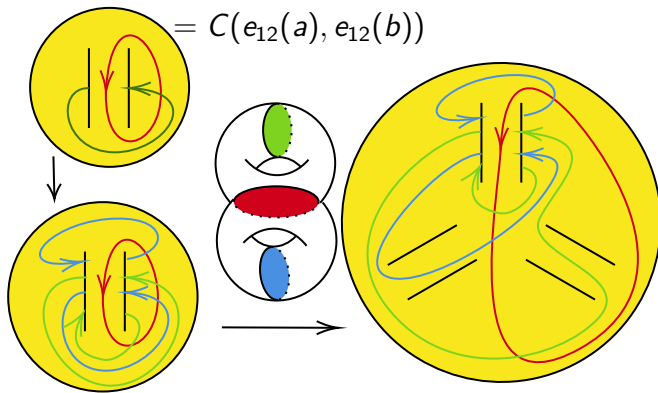
Notice $e_{ij}(a)^{-1} = e_{ij}(-a)$, and they satisfy the Steinberg relations:

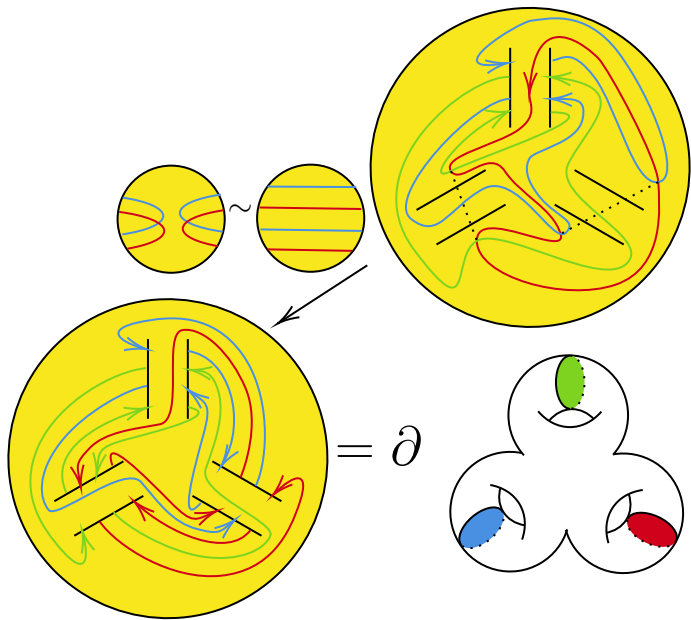
$$[e_{ij}(a), e_{kl}(b)] = \begin{cases} e_{il}(ab), & \text{if } j = k, \\ \mathbf{1}, & \text{otherwise.} \end{cases}$$

Using these relations and Hall-Witt identity, we have the following non-trivial prove of $[e_{12}(a), e_{12}(b)] = \mathbf{1}$:

$$\begin{aligned} [e_{12}(a), e_{12}(b)] &= [[e_{13}(a), e_{32}(1)], e_{12}(b)] \\ &= [e_{32}(1), [e_{13}(-a), e_{12}(-b)]]^{e_{13}(a)} [e_{13}(a), [e_{32}(-1), e_{12}(-b)]]^{e_{32}(-1)} \\ &= [e_{32}(1), \mathbf{1}]^{e_{13}(a)} [e_{13}(a), \mathbf{1}]^{e_{32}(-1)} = \mathbf{1} \end{aligned}$$

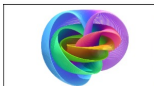
Following this blueprint, we have such handle decomposition to construct a lasagna:





Thank you, hope you have got the brain upgrades!

$$S^3 \sim \mathbb{C}^2 \setminus \{0\} \rightarrow \mathbb{CP}^1 \sim S^2$$



$$\pi_4(S^3) \cong \mathbb{Z}/2\mathbb{Z}$$

$$\pi_1^s(S) \cong \Omega_1^{\text{fr}}$$

$$K_1(\mathbb{F}_1) \cong S_\infty^{\text{ab}}$$

$$0=1-1=1+1=0$$

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